Nexus Platform - High-Level System Architecture Specification

Nexus Platform - High-Level System Architecture Specification

SECTION 1: ARCHITECTURE OVERVIEW & INGRESS TIER

The Nexus Cloud Platform is built on a distributed microservices architecture hosted across multiple Amazon Web Services (AWS) Availability Zones in us-west-2 (Oregon) with an active-standby disaster recovery presence in us-east-1. Ingress traffic is fronted by Cloudflare Enterprise for DDoS mitigation and SSL termination, flowing through an Envoy API Gateway cluster with mTLS service mesh routing.

SECTION 2: EVENT-DRIVEN MESSAGE BUS & SERVICES TIER

Core services communicate asynchronously via an Apache Kafka 3.6 event bus deployed with 3 brokers and a replication factor of 3. Core microservices include:

- Authentication & RBAC Service (FastAPI / Auth0 / Local JWT RS256/HS256)

- Ingestion Pipeline Engine (LlamaParse, async chunking, metadata enrichment)

- Vector Search & Reranking Service (Qdrant hybrid dense-sparse, BGE-Reranker-Large)

- LLM Grounding & Citation Synthesizer (LangChain, OpenAI GPT-4o, Google Gemini Flash)

- Chat History Persistence Service (PostgreSQL 16 with async SQLAlchemy)

SECTION 3: CACHING & PERSISTENCE TIERS

- Primary Relational Database: PostgreSQL 16 hosted on AWS Aurora with Citus multi-node distributed tables and monthly partitioning on chat history.

- Caching Layer: Redis Cluster 7.2 configured with write-through caching, 4-hour TTL for session state, and LRU eviction policy.

- Vector Engine: Qdrant cluster storing 384-dimensional dense vectors (BAAI/bge-small-en-v1.5) and SPLADE sparse lexical vectors, enabling Reciprocal Rank Fusion (RRF) search.

SECTION 4: OBSERVABILITY & DISASTER RECOVERY

Observability is maintained through OpenTelemetry instrumentation streaming to Datadog and Prometheus/Grafana. P99 API response latency is maintained below 320ms. The platform satisfies a Recovery Point Objective (RPO) of <1 minute and a Recovery Time Objective (RTO) of <15 minutes.